

Recursive Macroeconomic Theory

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Complete Solutions

Chapter 3 — Dynamic Programming

The single end-of-chapter exercise, 3.1 (p. 114)

Chapter 3 closes with one exercise, *Howard's policy iteration algorithm* on the stochastic Brock–Mirman model. It is analytical: it asks to show that the algorithm lands on the optimal policy in a single step. The solution below does every step of the algebra, and derives the closed form of the value of an arbitrary policy by two independent routes.

The companion notebook `ls_ch03.ipynb` (same folder) checks each analytical claim numerically — series, recursion and Monte Carlo for the policy value; a numerical maximisation with Gauss–Hermite quadrature for the improvement step; the Bellman residual — and then solves a discretised version by value iteration and policy iteration. Every check is an **assert**.

Notation follows the book: k capital, θ the productivity shock, h a policy, J_h the value of following it forever. Blue boxes are odd-numbered exercises.

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Chapter 3 Dynamic Programming

The chapter replaces the search for an infinite *sequence* of controls by the search for a pair of *functions*: a value function $V(x)$ and a policy $u = h(x)$ linked by the Bellman equation (3.1.5),

$$V(x) = \max_u \{r(x, u) + \beta V[g(x, u)]\}.$$

Three methods (section 3.1.1) solve it:

- *Value function iteration* (3.1.10): $V_{j+1}(x) = \max_u \{r(x, u) + \beta V_j[g(x, u)]\}$ from $V_0 = 0$. Converges at a linear (geometric) rate.
- *Guess and verify*: posit a functional form with undetermined coefficients, as in (3.1.12)–(3.1.14).
- *Howard's improvement algorithm*: (1) evaluate a policy, $V_{h_j}(x) = \sum_t \beta^t r[x_t, h_j(x_t)]$; (2) improve it, $h_{j+1} = \arg \max_u \{r(x, u) + \beta V_{h_j}[g(x, u)]\}$; (3) repeat. It is Newton's method on the Bellman equation (footnote 6), so it converges quadratically.

The benchmark. With ln utility and Cobb–Douglas technology, $k_{t+1} + c_t = Ak_t^\alpha$, the chapter finds (p. 110)

$$v(k) = E + F \ln k, \quad F = \frac{\alpha}{1 - \alpha\beta}, \quad \tilde{k} = \beta\alpha Ak^\alpha,$$

$$E = \frac{1}{1 - \beta} \left[\ln A(1 - \alpha\beta) + \frac{\beta\alpha}{1 - \alpha\beta} \ln A\beta\alpha \right].$$

Exercise 3.1 adds an i.i.d. shock and shows that, in this model, Howard's algorithm needs only one improvement step.

Policy Iteration in the Brock–Mirman Model

Exercise 3.1 Howard's policy iteration algorithm

Question

Consider the Brock-Mirman problem: to maximize

$$E_0 \sum_{t=0}^{\infty} \beta^t \ln c_t,$$

subject to $c_t + k_{t+1} \leq Ak_t^\alpha \theta_t$, k_0 given, $A > 0$, $1 > \alpha > 0$, where $\{\theta_t\}$ is an i.i.d. sequence with $\ln \theta_t$ distributed according to a normal distribution with mean zero and variance σ^2 .

Consider the following algorithm. Guess at a policy of the form $k_{t+1} = h_0(Ak_t^\alpha \theta_t)$ for any constant $h_0 \in (0, 1)$. Then form

$$J_0(k_0, \theta_0) = E_0 \sum_{t=0}^{\infty} \beta^t \ln (Ak_t^\alpha \theta_t - h_0 Ak_t^\alpha \theta_t).$$

Next choose a new policy h_1 by maximizing

$$\ln (Ak^\alpha \theta - k') + \beta E J_0(k', \theta'),$$

where $k' = h_1 Ak^\alpha \theta$. Then form

$$J_1(k_0, \theta_0) = E_0 \sum_{t=0}^{\infty} \beta^t \ln (Ak_t^\alpha \theta_t - h_1 Ak_t^\alpha \theta_t).$$

Continue iterating on this scheme until successive h_j have converged.

Show that, for the present example, this algorithm converges to the optimal policy function in one step.

Solution:

Throughout, write output as $y_t \equiv Ak_t^\alpha \theta_t$. The constraint binds (utility is strictly increasing in c), so $c_t = y_t - k_{t+1}$. We need a formula for the value J_h of an *arbitrary* constant savings rate $h \in (0, 1)$, then the improvement step, then a check that the improved policy cannot be improved further.

Step 1 — the value of the policy $k_{t+1} = hy_t$.

Under this policy $c_t = (1 - h)y_t$, so $\ln c_t = \ln(1 - h) + \ln y_t$. The law of motion of $\ln y$ is

$$\begin{aligned} \ln y_{t+1} &= \ln A + \alpha \ln k_{t+1} + \ln \theta_{t+1} = \ln A + \alpha \ln(hy_t) + \ln \theta_{t+1} \\ &= (\ln A + \alpha \ln h) + \alpha \ln y_t + \ln \theta_{t+1}, \end{aligned} \tag{1}$$

an AR(1) with coefficient $\alpha \in (0, 1)$ and i.i.d. innovation of mean $E \ln \theta = 0$. Let $\mu_h \equiv \ln A + \alpha \ln h$.

Route 1: sum the series. Iterating (1) back to $t = 0$,

$$\ln y_t = \alpha^t \ln y_0 + \mu_h \sum_{s=0}^{t-1} \alpha^s + \sum_{s=1}^t \alpha^{t-s} \ln \theta_s, \quad \text{so} \quad E_0 \ln y_t = \alpha^t \ln y_0 + \mu_h \frac{1 - \alpha^t}{1 - \alpha},$$

because $E_0 \ln \theta_s = 0$ for $s \geq 1$. Every term has finite mean and the discounted sum converges absolutely, so expectation and sum can be interchanged:

$$\begin{aligned} J_h(k_0, \theta_0) &= \sum_{t=0}^{\infty} \beta^t [\ln(1-h) + E_0 \ln y_t] \\ &= \frac{\ln(1-h)}{1-\beta} + \ln y_0 \sum_{t=0}^{\infty} (\alpha\beta)^t + \frac{\mu_h}{1-\alpha} \left[\sum_{t=0}^{\infty} \beta^t - \sum_{t=0}^{\infty} (\alpha\beta)^t \right] \\ &= \frac{\ln(1-h)}{1-\beta} + \frac{\ln y_0}{1-\alpha\beta} + \frac{\mu_h}{1-\alpha} \left[\frac{1}{1-\beta} - \frac{1}{1-\alpha\beta} \right]. \end{aligned}$$

The bracket is $\frac{(1-\alpha\beta) - (1-\beta)}{(1-\beta)(1-\alpha\beta)} = \frac{\beta(1-\alpha)}{(1-\beta)(1-\alpha\beta)}$, and the $(1-\alpha)$ cancels. Hence

$$J_h(k, \theta) = \underbrace{\frac{\ln(1-h)}{1-\beta} + \frac{\beta(\ln A + \alpha \ln h)}{(1-\beta)(1-\alpha\beta)}}_{\equiv a(h)} + \frac{1}{1-\alpha\beta} \ln(Ak^\alpha \theta) \quad (2)$$

For $h \in (0, 1)$ both $\ln h$ and $\ln(1-h)$ are finite, so J_h is finite for every (k, θ) with $k > 0$.

Route 2: undetermined coefficients. J_h satisfies the policy-evaluation recursion $J_h(k, \theta) = \ln[(1-h)y] + \beta E J_h(hy, \theta')$. Guess $J_h = a + b \ln y$. Using (1) for next period's $\ln y'$,

$$\begin{aligned} a + b \ln y &= \ln(1-h) + \ln y + \beta [a + b(\mu_h + \alpha \ln y + E \ln \theta')] \\ &= \ln(1-h) + \beta a + \beta b \mu_h + (1 + \alpha\beta b) \ln y. \end{aligned}$$

Matching the coefficient on $\ln y$: $b = 1 + \alpha\beta b \Rightarrow b = \frac{1}{1-\alpha\beta}$. Matching constants:

$$(1-\beta)a = \ln(1-h) + \frac{\beta\mu_h}{1-\alpha\beta}, \text{ i.e. } a = \frac{\ln(1-h)}{1-\beta} + \frac{\beta(\ln A + \alpha \ln h)}{(1-\beta)(1-\alpha\beta)} \text{ — the same } a(h)$$

as Route 1. ✓

The key feature. Expanding $\ln y = \ln A + \alpha \ln k + \ln \theta$ in (2),

$$J_h(k, \theta) = a(h) + \frac{\ln A + \ln \theta}{1-\alpha\beta} + \frac{\alpha}{1-\alpha\beta} \ln k. \quad (3)$$

The policy h enters *only the constant*. The coefficient on $\ln k$ — the elasticity of lifetime utility with respect to capital — is $\alpha/(1-\alpha\beta)$ for *every* h .

Step 2 — the improvement step.

Take $J_0 = J_{h_0}$ for any $h_0 \in (0, 1)$. By (3) and $E \ln \theta' = 0$,

$$E J_0(k', \theta') = a(h_0) + \frac{\ln A}{1 - \alpha\beta} + \frac{\alpha}{1 - \alpha\beta} \ln k' \equiv \kappa(h_0) + \frac{\alpha}{1 - \alpha\beta} \ln k'.$$

The improvement problem is therefore

$$\max_{0 < k' < y} \ln(y - k') + \beta \kappa(h_0) + \frac{\alpha\beta}{1 - \alpha\beta} \ln k'.$$

The objective is strictly concave in k' (a sum of logs of affine functions) and tends to $-\infty$ at both ends of $(0, y)$, so the first-order condition identifies the unique maximum:

$$\frac{1}{y - k'} = \frac{\alpha\beta}{(1 - \alpha\beta)k'} \implies (1 - \alpha\beta)k' = \alpha\beta(y - k') \implies k' = \alpha\beta y.$$

So

$$\boxed{h_1 = \alpha\beta \text{ for every } h_0 \in (0, 1)}$$

— independently of h_0 , of the state (k, θ) , and of σ^2 . The constant $\kappa(h_0)$, the only place h_0 appears, drops out of the first-order condition.

Step 3 — h_1 is a fixed point, and it is the optimum.

The algorithm stops. $J_1 = J_{\alpha\beta}$ is again of the form (3), so repeating Step 2 with h_0 replaced by h_1 gives $h_2 = \alpha\beta = h_1$. Successive h_j have converged after one improvement. *J_1 solves the Bellman equation.* Let $(TJ)(k, \theta) \equiv \max_{k'} \{\ln(y - k') + \beta EJ(k', \theta')\}$. Step 2 applied to J_1 says the maximiser is $k' = \alpha\beta y$, so

$$(TJ_1)(k, \theta) = \ln[(1 - \alpha\beta)y] + \beta EJ_1(\alpha\beta y, \theta') = J_1(k, \theta),$$

where the last equality is the policy-evaluation recursion of Route 2 at $h = \alpha\beta$. Hence $J_1 = TJ_1$: J_1 and h_1 solve the functional equation (3.1.5)–(3.1.6), which has a unique solution (finding 1, p. 107). Therefore $J_1 = V$ and $h_1 = \alpha\beta$ is the optimal policy, $k_{t+1} = \alpha\beta A k_t^\alpha \theta_t$. ■

Explicit form, and a check against the chapter. Setting $h = \alpha\beta$ in (2),

$$V(k, \theta) = \frac{\ln(1 - \alpha\beta)}{1 - \beta} + \frac{\beta \ln A + \alpha\beta \ln(\alpha\beta)}{(1 - \beta)(1 - \alpha\beta)} + \frac{\ln A}{1 - \alpha\beta} + \frac{\alpha}{1 - \alpha\beta} \ln k + \frac{\ln \theta}{1 - \alpha\beta}.$$

The two $\ln A$ terms combine to $\ln A \frac{\beta + (1 - \beta)}{(1 - \beta)(1 - \alpha\beta)} = \frac{\ln A}{(1 - \beta)(1 - \alpha\beta)}$. The book's

deterministic constant expands to $E = \frac{1}{1 - \beta} \left[\ln A + \ln(1 - \alpha\beta) + \frac{\alpha\beta(\ln A + \ln \alpha\beta)}{1 - \alpha\beta} \right]$,

whose $\ln A$ terms are $\frac{\ln A}{1 - \beta} \left[1 + \frac{\alpha\beta}{1 - \alpha\beta} \right] = \frac{\ln A}{(1 - \beta)(1 - \alpha\beta)}$, and whose remaining terms

$\frac{\ln(1 - \alpha\beta)}{1 - \beta} + \frac{\alpha\beta \ln \alpha\beta}{(1 - \beta)(1 - \alpha\beta)}$ match ours term by term. So

$$V(k, \theta) = E + F \ln k + \frac{\ln \theta}{1 - \alpha\beta}, \quad F = \frac{\alpha}{1 - \alpha\beta},$$

the chapter's (3.1.12)–(3.1.14) plus the current shock. At $\theta = 1$ it is exactly the deterministic solution: with log utility the shock variance σ^2 changes neither the policy nor the value. ✓

Contrast — value iteration from $v_0 = 0$. Guess $v_j = e_j + f_j \ln y$ with $f_0 = 0$. Then $Ev_j(k', \theta') = e_j + f_j(\ln A + \alpha \ln k')$ and the first-order condition of $\max_{k'} \{\ln(y - k') + \beta Ev_j\}$ is $\frac{1}{y - k'} = \frac{\alpha\beta f_j}{k'}$, so $k' = \frac{\alpha\beta f_j}{1 + \alpha\beta f_j} y$. Substituting, $\ln(y - k') = \ln y - \ln(1 + \alpha\beta f_j)$ and $\ln k' = \ln y + \text{const}$, so the coefficient on $\ln y$ in v_{j+1} is $f_{j+1} = 1 + \alpha\beta f_j$, giving $f_j = \frac{1 - (\alpha\beta)^j}{1 - \alpha\beta}$. The policy produced when computing v_{j+1} is

$$h_{j+1}^{VFI} = \frac{\alpha\beta f_j}{1 + \alpha\beta f_j} = \alpha\beta \frac{1 - (\alpha\beta)^j}{1 - (\alpha\beta)^{j+1}}, \quad h_{j+1}^{VFI} - \alpha\beta = -\frac{(\alpha\beta)^{j+1}(1 - \alpha\beta)}{1 - (\alpha\beta)^{j+1}}.$$

(For the gap: $\frac{1 - (\alpha\beta)^j}{1 - (\alpha\beta)^{j+1}} - 1 = \frac{(\alpha\beta)^{j+1} - (\alpha\beta)^j}{1 - (\alpha\beta)^{j+1}}$, times $\alpha\beta$.) Value iteration approaches $\alpha\beta$ only geometrically, at rate $\alpha\beta$, and never reaches it; Howard's algorithm reaches it exactly after one step. This is the example the chapter cites (p. 109) for the claim that policy iteration often converges faster.

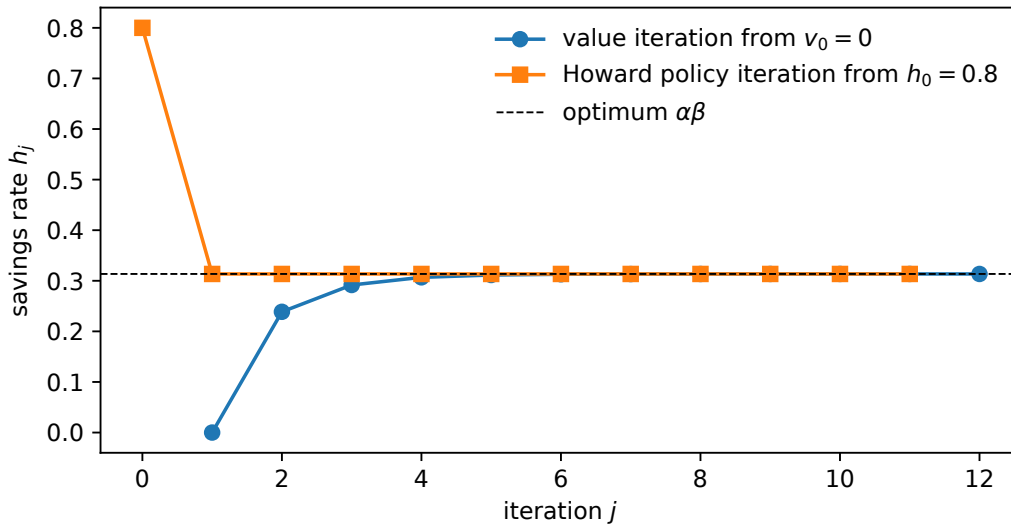


Figure 1: Exercise 3.1. Savings rate h_j by iteration, $\alpha = 0.33$, $\beta = 0.95$ ($\alpha\beta = 0.3135$). Howard from $h_0 = 0.8$ lands on $\alpha\beta$ at $j = 1$; value iteration from $v_0 = 0$ gives 0, 0.2387, 0.2917, 0.3068, 0.3114, ...

The improvement step uses J_{h_0} only through its *derivative* in k' — the marginal value of capital. Changing the savings rate rescales consumption in every period and state by the same factor $(1 - h)$, which in logs is an additive constant; and it shifts the mean of $\ln k$ by a constant. Neither changes how lifetime utility responds to $\ln k$, which is $\alpha(1 + \alpha\beta + (\alpha\beta)^2 + \dots) = \alpha/(1 - \alpha\beta)$ for every h : a unit of $\ln k$ today raises $\ln y$ by α , and that effect persists at rate α and is discounted at β . Since the *wrong* policy already carries the *right* marginal value of capital, the first improvement is already optimal. Contrast value iteration: there the slope f_j is built up one term of the geometric series per iteration, and the policy inherits the error.

Numerical exercise — see the companion code

Notebook `ls_ch03.ipynb`, with $A = 1$, $\alpha = 0.33$, $\beta = 0.95$, $\sigma = 0.2$ (the book gives no values).

- (2) agrees with the exact mean recursion to 10^{-9} and with a 20 000-path Monte Carlo within 5 standard errors, for $h_0 \in \{0.1, 0.5, 0.9\}$.
- Numerical improvement step (bounded scalar maximisation, E by 20-node Gauss–Hermite): $h_1 \in [0.31349997, 0.31350002]$ for $h_0 \in \{0.05, 0.2, 0.5, 0.8, 0.95\}$ and nine states each; $\alpha\beta = 0.3135$. A second step returns the same value.
- Bellman residual $\max |TJ_{\alpha\beta} - J_{\alpha\beta}| = 7 \times 10^{-15}$ on a grid of states; for $h_0 = 0.5$, $TJ_{0.5} - J_{0.5} = 0.104 > 0$ (strict improvement).
- The value-iteration sequence h_j^{VFI} above matches a numerical Bellman iteration to 10^{-8} .

Discretised model — see the companion notebook

Capital on a 250-point geometric grid over $[0.2\bar{k}, 4\bar{k}]$, $\ln \theta$ on 7 Gauss–Hermite nodes (symmetric, so $E \ln \theta' = 0$ and the closed form stays the exact benchmark). Value iteration needs **450** iterations to reach a sup-norm change below 10^{-10} ; Howard, started from $h_0 = 0.25$ and evaluating each policy by solving $V = R_h + \beta P_h V$, stops after **5** improvements plus one confirming step. Both give the identical grid policy, with $k'/y \in [0.3114, 0.3156]$ on interior states and $\max |V_{\text{grid}} - V| = 8.9 \times 10^{-5}$.

On the grid, one step is *not* enough: the first improvement leaves $\max |k'/y - \alpha\beta| = 0.0125$, then 0.0058, 0.0028, 0.0022, 0.0021 (the grid-spacing floor). Rounding $h_0 y$ to the grid breaks the exact form (3), which the one-step argument relies on.

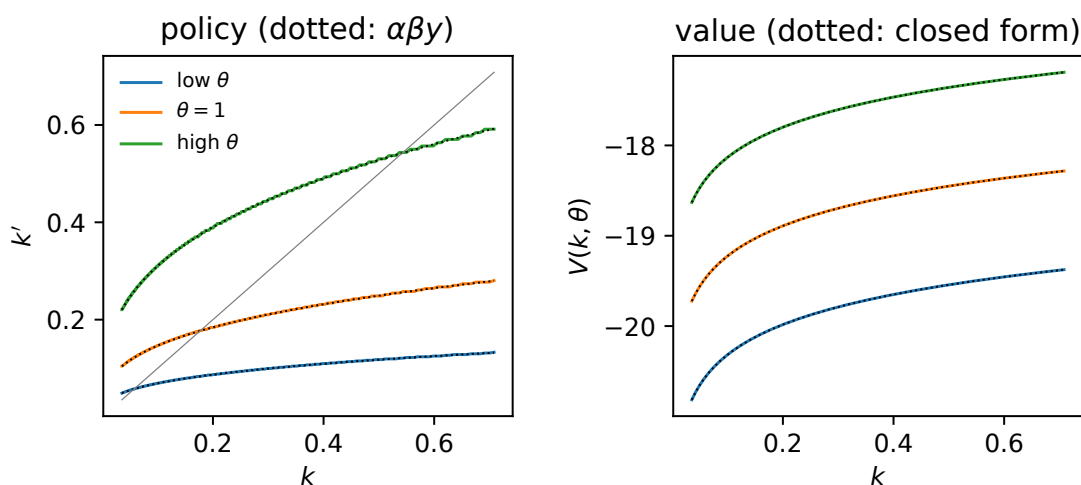


Figure 2: Exercise 3.1, discretised model. Left: grid policy for three shock values against $\alpha\beta y$ (dotted). Right: grid value function against the closed form $E + F \ln k + \ln \theta / (1 - \alpha\beta)$ (dotted).

Where this goes wrong

- **$E\theta$ is not 1.** With $\ln \theta \sim N(0, \sigma^2)$, $E\theta = e^{\sigma^2/2}$. What the derivation needs is $E \ln \theta' = 0$, and only $\ln \theta$ ever enters because utility is logarithmic and technology Cobb–Douglas. Using $E\theta = 1$ inside a log is Jensen’s error.
- **Improving over the wrong variable.** The maximisation is over k' with J_0 held fixed; do not substitute $k' = h_1 y$ into J_0 ’s own h_0 . The new policy enters only through the choice of k' today; tomorrow onward the old policy h_0 is followed.
- **Forgetting the h -dependence of the drift.** $\mu_h = \ln A + \alpha \ln h$ depends on h ; it matters for the level $a(h)$ and the comparison $J_{\alpha\beta} \geq J_h$, even though it does not affect h_1 .
- **Over-generalising.** One-step convergence is special to the log / Cobb–Douglas / full-depreciation case, and even there only in the continuous state space: the discretised version above needs five improvements.

Companion code

Every numerical result above is reproduced by the notebook `ls_ch03.ipynb` in this folder (executed, outputs saved). Each claim is checked by an independent route with an `assert`.

Notebook section	What it does
Step 1	Closed form (2) vs exact mean recursion and Monte Carlo.
Step 2	Numerical improvement step from five values of h_0 ; $h_1 = h_2 = \alpha\beta$.
Step 3	Bellman residual of $J_{\alpha\beta}$; strict improvement for $h \neq \alpha\beta$.
Contrast	Value-iteration savings rates, closed form vs numerical; Figure 1.
Discretised	Grid VFI vs grid PFI (identical policies and values), comparison with the closed form; Figure 2.

Figures are saved to `fig/ls03_*.pdf` with TrueType (not Type 3) fonts.